MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPES LSTCJX, LSTCKX AND LSTCTX

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Stranded conductor, uncoated (see table I for size and stranding).
Second - Extruded silicone rubber insulation (see table I for dimensions).
Third - Glass braid, color identified (see table II). (See table I for dimensions).
Fourth - Braid covering. Two conductors cabled to form a pair, with lay not greater than 2.25 inches. For LSTCJX, one iron and one constantan conductor. For LSTCKX, one chromel and one alumel conductor. For LSTCTX, one copper and one constantan conductor. Type J, K and T thermocouple conductors shall be in accordance with ANSI MC96.1.
Fifth - The specified number of pairs (see table I) cabled together with a lay not greater than 10 inches. Fillers saturated with noncharring silicone-base compound shall be used as necessary to make a firm, round cable.
Sixth - Binder tape applied helically with overlap.
Seventh - Silicone rubber jacket (see table I for thickness).
Eighth - Braided metal armor.
Ninth - Paint.

FSC 6145

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TABLE I. Details.

Military part no. M24643/24	Cable type and size	Conductors				Cable jacket thickness minimum (inch)	Overall diameter maximum (inch)	Cold bending mandrel diameter maximum (inches)
		Number of pairs	Number of strands and strand diameter (inch)	Diameter over insulation minimum (inch)	Diameter over braid maximum (inch)			
-01AN	LSTCJX-3	3	7/0.0201	0.100	0.125	0.040	0.742	9.0
-02AN	LSTCJX-7	7	7/0.0201	.100	.125	.050	.983	11.0
-03AN	LSTCJX-12	12	7/0.0201	.100	.125	.050	1.269	15.0
-04AN	LSTCKX-1	1	7/0.0201	.100	.125	.030	.456	5.0
-05AN	LSTCKX-3	3	7/0.0201	.100	.125	.040	.742	9.0
-06AN	LSTCKX-7	7	7/0.0201	.100	.125	.050	.983	11.0
-07AN	LSTCKX-12	12	7/0.0201	.100	.125	.050	1.269	15.0
-08AN	LSTCTX-1	1	7/0.0113	.065	.085	.030	.350	4.25
-09AN	LSTCTX-3	3	7/0.0113	.065	.085	.030	.552	6.5
-10AN	LSTCTX-7	7	7/0.0113	.065	.085	.040	.731	8.0
-11AN	LSTCTX-12	12	7/0.0113	.065	.085	.050	.964	11.0

TABLE II. Color identification code.

Pair number	Constantan or alumel	Iron, copper or chromel	
		Base color	Tracer color
1	Red	White	———
2	Red	Black	———
3	Red	Green	———
4	Red	Orange	———
5	Red	Blue	———
6	Red	White	Black
7	Red	White	Green
8	Red	White	Orange
9	Red	White	Blue
10	Red	White	Red
11	Red	Black	White
12	Red	Green	White

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
ANSI MC96.1 Calibration limits of error:	
Type J	± 2.2°C or 0.75 percent
Type K:	
minus 32 to 530°F	± 4°F
532 to 2300°F	± 0.75 percent
Type T	± 1°C or 0.75 percent
Voltage withstand - volts, root mean square, minimum	2000
Insulation resistance - megohms/1000 feet, minimum	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Group B:	
Cold bending, cable (see table I for mandrel size)	No damage
Gas flame, 1 hour at 120/70 volt root mean square	No failure
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	800
Elongation - percent, minimum	250
Durometer (type A) hardness, minimum (ASTM D 2240)	60
Group C:	
Physicals - cable aging (260 ± 10°C)	
Jacket	
Tensile strength - lb/in^2, minimum	600
Elongation - percent, minimum	150

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (125 ± 5°C)	No failure
Gas flame (3 hours) at 120/70 volts root mean square	No failure
Armor - conformance to material, construction, and coverage	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Flame propagation	No failure

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UNIT ORDERING LENGTHS:
All types and sizes 100 feet minimum

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
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